Na poprzednich zajęciach:

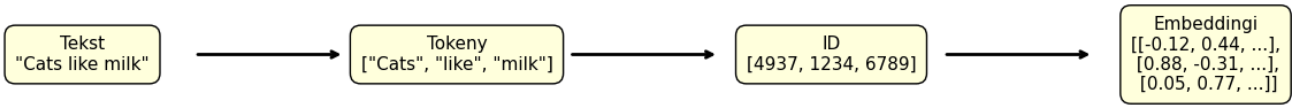
- **tokenizacja** – podział tekstu na słowa / tokeny,
- **spaCy Matcher** – znajdowanie wzorców na podstawie części mowy i cech morfologicznych.

Wektory słów

Tokenizacja a wektory słów

Tokenizacja → Podziel tekst na kawałki (ID ze słownika)

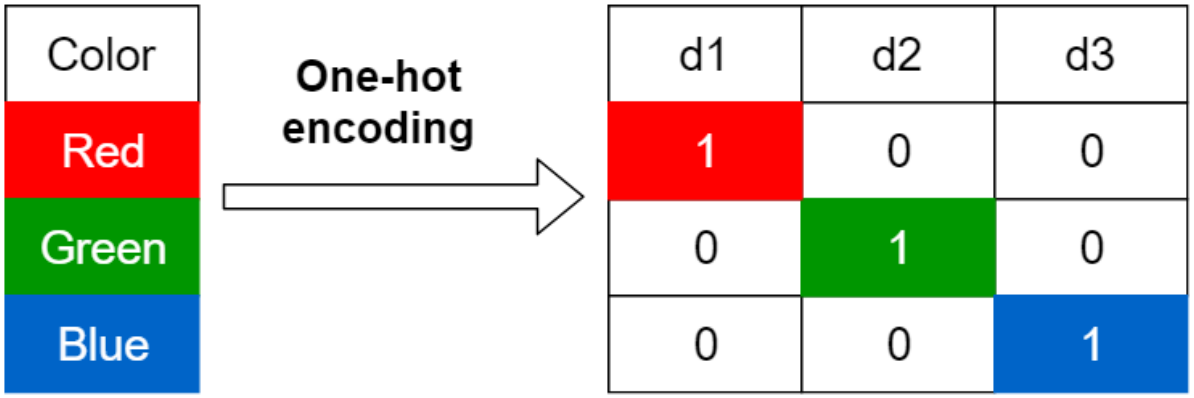
Embedding → Przedstaw każde słowo w postaci liczb, aby algorytm mógł z nim dalej pracować



Komputer nie „rozumie" tekstu, tylko operuje na liczbach. Zatem każde słowo (np. „król", „kot", „miłość") trzeba jakoś zamienić na liczby, żeby można było nim manipulować w modelu.

Najprostszy pomysł — **one-hot encoding**

```
słownik = [kot, król, królowa]
"król" → [0, 1, 0]
```



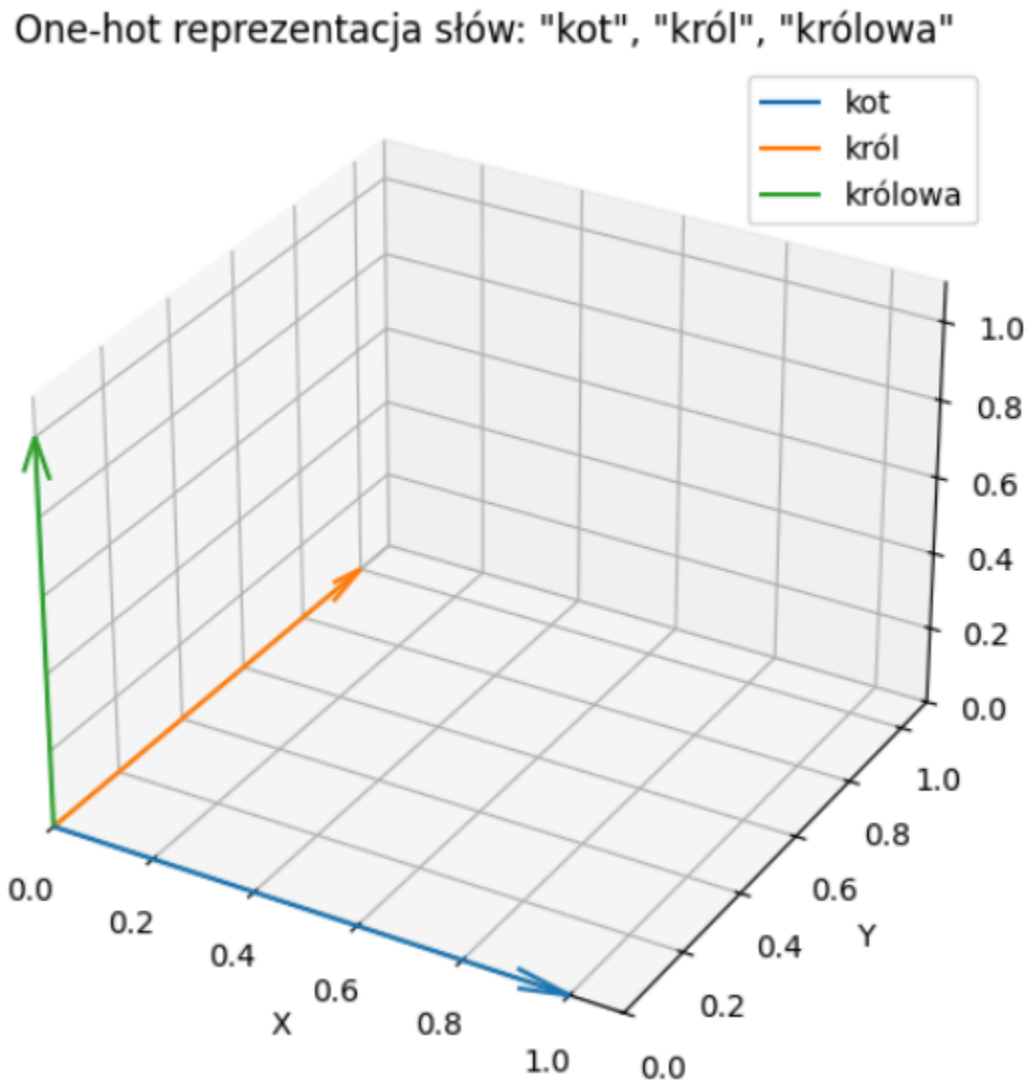
Dlaczego one-hot encoding jest problematyczny?

❌ Bardzo duży wymiar - Jeśli mamy słownik np. 100 000 słów, każdy wektor ma długość 100 000 — to ogromne marnotrawstwo pamięci.

❌ Brak informacji o znaczeniu - Wszystkie wektory są od siebie tak samo odległe — np. "kot" i "pies" są tak samo różne jak "kot" i "samolot". Nie ma żadnego pojęcia o podobieństwie semantycznym (znaczeniowym).

❌ Brak możliwości uogólniania - Model nie może „zrozumieć”, że pewne słowa mają podobne znaczenie — każde słowo jest całkowicie niezależne.

```
słownik = [kot, król, królowa]
"kot" → [1, 0, 0]
"król" → [0, 1, 0]
"królowa" → [0, 0, 1]
```



Widzimy, że każdy wektor idzie w swoją stronę, każdy jest tak samo odległy od siebie (90 stopni). Żadne dwa nie są do siebie bliżej ani dalej — odległość między każdym z nich jest taka sama.

W naszym przykładzie: kot, król i królowa są tak samo podobni i różni = nie ma wartości semantycznej.

Embedingi

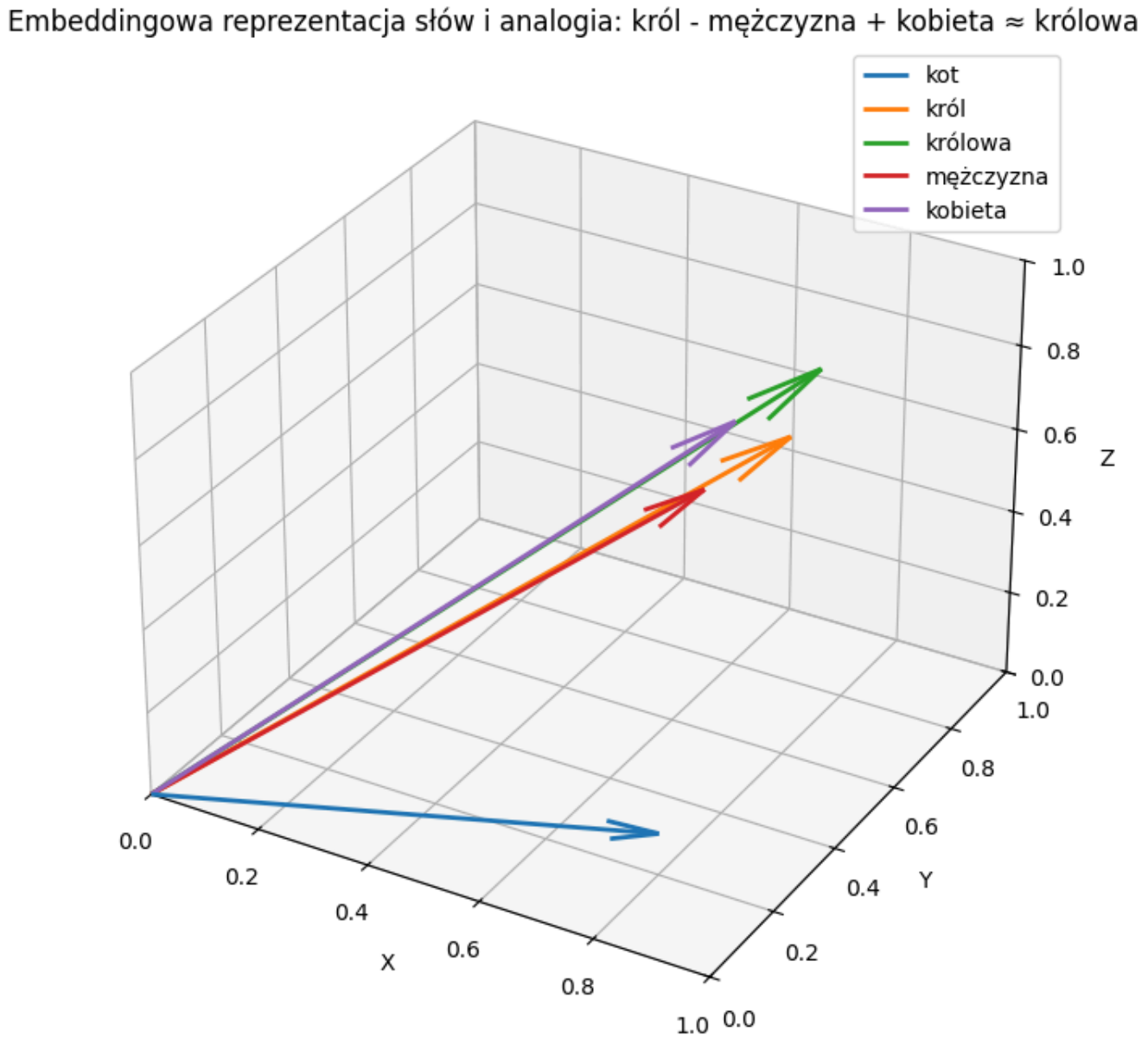
Embeddingi rozwiązują ten problem. W embeddingach uczymy model, żeby słowa, które występują w podobnych kontekstach, miały **podobne wektory**.

Dwa wektory są podobne, kiedy wskazują w podobnym kierunku w przestrzeni wielowymiarowej. Czyli: jeśli narysujesz te wektory jako strzałki zaczynające się w tym samym

punkcie (np. w punkcie (0,0,0)) to mają **mały kąt między sobą**.

Przykład:

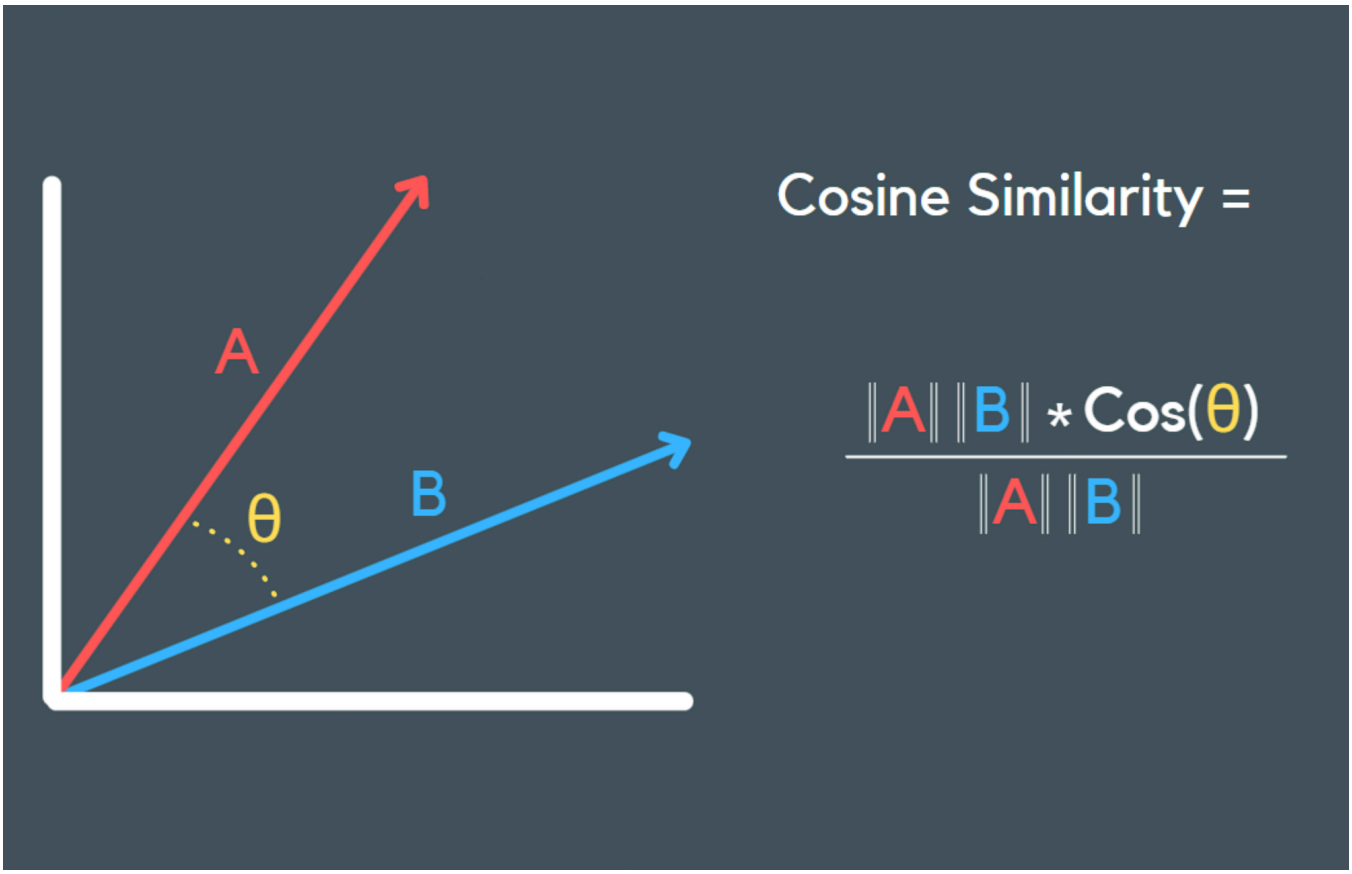
```
"king" → [0.25, 0.88, -0.12, 0.40, ...]
"queen" → [0.22, 0.90, -0.10, 0.45, ...]
"cat" → [-0.70, 0.11, 0.45, 0.20, ...]
```




```
król - mężczyzna + kobieta ≈ królowa
```

Teraz punkty w przestrzeni układają się tak, że słowa o podobnym znaczeniu są blisko siebie. To jest właśnie **semantyczne podobieństwo** — podobieństwo w sensie znaczenia, a nie liter czy indeksów w tabeli.

Cosinusowe podobieństwo — jak mierzymy „bliskość znaczeń”



Cosinusowe podobieństwo to sposób na sprawdzenie, czy dwa wektory są „w tym samym kierunku”:

- 1 → bardzo podobne (np. „kot” i „pies”)
- 0 → brak podobieństwa (np. „kot” i „okno”)
- -1 → całkowite przeciwieństwo

Ważne: Patrzymy nie na długość wektorów, tylko na kąt między nimi. Dwa wektory, które „patrzą” w tym samym kierunku, oznaczają słowa o podobnym znaczeniu.

Skąd się biorą te wektory?

Z treningu modelu językowego (np. Word2Vec, GloVe, FastText, BERT). Model uczy się na dużych korpusach tekstu, obserwując konteksty słów.

Efekt:

„king” i „queen” często występują w podobnych zdaniach (np. royal palace, throne, reign) → mają podobne wektory.

„dog” i „cat” też → inna grupa semantyczna.

Trochę praktyki

Zaczniemy od zapoznania się z biblioteką **gensim** Jej nazwa pochodzi od "GENerate SIMilar" — bo jej głównym celem jest znajdowanie podobieństwa między dokumentami lub słowami.

Użyjemy klasy **KeyedVectors** - to strukturą danych w gensim, która przechowuje wektory słów (embeddingi) wytrenowane dowolnym modelem. Czyli po prostu taki kontener do pracy z wektorami - niezależnie od tego, jak one zostały wytrenowane.

In [2]:
```
!pip install gensim
```

```
Collecting gensim
  Downloading gensim-4.4.0-cp313-cp313-
manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (8.4 kB)
Requirement already satisfied: numpy⩾1.18.5 in
/usr/local/lib/python3.13/dist-packages (from gensim) (2.1.3)
Requirement already satisfied: scipy⩾1.7.0 in
/usr/local/lib/python3.13/dist-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open⩾1.8.1 in
/usr/local/lib/python3.13/dist-packages (from gensim) (8.0.2)
Requirement already satisfied: wrapt in /usr/local/lib/python3.13/dist-
packages (from smart_open⩾1.8.1→gensim) (2.5.0)
Downloading gensim-4.4.0-cp313-cp313-
manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (27.8 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 27.8/27.8 MB 78.1 MB/s eta
0:00:00
Installing collected packages: gensim
Successfully installed gensim-4.4.0
```

In [3]:
```
import gensim
from gensim.models import KeyedVectors
```

weźmiemy sobie polskie pre-trenowane wektory: http://dsmodels.nlp.ipipan.waw.pl pobieramy plik:

In [4]:
```
!wget http://dsmodels.nlp.ipipan.waw.pl/dsmodels/nkjp+wiki-forms-restricted-300-skipg-ns.txt.gz --no-check-certificate
```

```
--2026-10-12 18:09:07--
http://dsmodels.nlp.ipipan.waw.pl/dsmodels/nkjp+wiki-forms-restricted-300-skipg-ns.txt.gz
Resolving dsmodels.nlp.ipipan.waw.pl (dsmodels.nlp.ipipan.waw.pl)...
213.135.36.94
Connecting to dsmodels.nlp.ipipan.waw.pl
(dsmodels.nlp.ipipan.waw.pl)|213.135.36.94|:80... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://dsmodels.nlp.ipipan.waw.pl/dsmodels/nkjp+wiki-forms-restricted-300-skipg-ns.txt.gz [following]
--2026-10-12 18:09:08--
https://dsmodels.nlp.ipipan.waw.pl/dsmodels/nkjp+wiki-forms-restricted-300-skipg-ns.txt.gz
Connecting to dsmodels.nlp.ipipan.waw.pl
(dsmodels.nlp.ipipan.waw.pl)|213.135.36.94|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2152612134 (2.0G) [application/octet-stream]
Saving to: ‘nkjp+wiki-forms-restricted-300-skipg-ns.txt.gz’

nkjp+wiki-forms-res 100%[===================>]   2.00G  18.2MB/s    in 2m
31s

2026-10-12 18:11:40 (13.6 MB/s) - ‘nkjp+wiki-forms-restricted-300-skipg-
ns.txt.gz’ saved [2152612134/2152612134]
```

Rozpakowujemy dane

In [5]:
```
!gzip -d "nkjp+wiki-forms-restricted-300-skipg-ns.txt.gz"
```

Badamy format pliku

In [ ]:
```
!head -n 100 nkjp+wiki-forms-restricted-300-skipg-ns.txt
#Pierwsza linia - lość słów (1982479), rozmiar wektora(300)
#Każda kolejna linia - słowo, liczby reperezentujące wektor
```

(to może chwilę zająć)

In [6]:
```
#Ładowanie wektorów z pliku do naszego KeyedVectors
wv = KeyedVectors.load_word2vec_format('nkjp+wiki-forms-restricted-300-skipg-ns.txt')
```

In [8]:
```
wv.key_to_index.keys()
wv['dom']
```

Out[8]:
```
array([-0.39294 , -0.156333,  0.017816, -0.208752,  0.442839, -0.389071,
       -0.22354 ,  0.119761,  0.201861,  0.136966,  0.005829, -0.036992,
       -0.075693, -0.424958,  0.031507, -0.214998, -0.11862 , -0.059799,
       -0.150971, -0.176567,  0.444136,  0.18925 ,  0.241265, -0.248733,
       -0.176206,  0.579075,  0.031083,  0.253453,  0.101754,  0.144021,
       -0.169662, -0.005122,  0.056225,  0.078038, -0.557848, -0.365922,
       -0.108863,  0.322074, -0.178232,  0.289982, -0.205699,  0.001991,
        0.001029, -0.074944,  0.173145,  0.044346, -0.032673, -0.060124,
        0.152098, -0.293121, -0.280878,  0.377744,  0.402627, -0.737609,
        0.020663, -0.01301 ,  0.134813, -0.052369,  0.046645, -0.024955,
        0.100392, -0.295437, -0.418354,  0.396864,  0.232666,  0.022819,
       -0.018474,  0.385309, -0.084371,  0.269105,  0.149824, -0.161652,
        0.282115,  0.177849, -0.241524, -0.240634, -0.111456,  0.070668,
       -0.216499, -0.139156, -0.576622, -0.136423,  0.403575,  0.155968,
        0.218948, -0.002413,  0.123077,  0.102393,  0.741444,  0.390994,
       -0.110964,  0.134329,  0.265171,  0.251443, -0.221384, -0.344603,
        0.283788,  0.284915,  0.057016,  0.384028, -0.733822,  0.303532,
       -0.583847, -0.02494 , -0.218198,  0.018208,  0.145667,  0.255729,
        0.498478, -0.112536, -0.047852, -0.337887,  0.031235, -0.432209,
        0.208977,  0.012221,  0.19098 , -0.206233, -0.113016,  0.111511,
       -0.059505,  0.027593, -0.100111, -0.230436, -0.078137,  0.278976,
        0.19794 ,  0.210459, -0.155359, -0.112397,  0.26439 , -0.199999,
        0.058368,  0.032107,  0.110054,  0.051876, -0.240713,  0.277577,
        0.289985, -0.200094, -0.101406,  0.131032, -0.059501,  0.134268,
        0.148928, -0.085373,  0.030669,  0.068627, -0.215063, -0.081981,
       -0.052877, -0.291941, -0.458875,  0.050618,  0.214273,  0.184849,
       -0.038208,  0.036236, -0.642125, -0.464941,  0.372085, -0.187011,
       -0.069729,  0.035051,  0.099067,  0.057524,  0.006579,  0.533439,
       -0.198236, -0.344194, -0.067796, -0.063289, -0.069952, -0.041706,
        0.142563,  0.05342 , -0.054871,  0.2577  ,  0.50673 ,  0.208496,
        0.650207, -0.252643,  0.320542, -0.54734 , -0.637634, -0.51038 ,
        0.025401,  0.05803 ,  0.16656 ,  0.257388,  0.305958, -0.219199,
        0.2086  , -0.251887, -0.185647, -0.027838, -0.186104, -0.070295,
       -0.094713,  0.102284, -0.123802,  0.083343,  0.315851, -0.031123,
        0.238093,  0.149717, -0.004316, -0.181932, -0.086956,  0.531083,
        0.112422,  0.231151,  0.242227,  0.012977, -0.095725, -0.283862,
       -0.061226,  0.311026,  0.146489,  0.054716, -0.048658, -0.282848,
       -0.054836, -0.030167, -0.090273, -0.435833, -0.287074, -0.351567,
       -0.002671, -0.020709, -0.276286,  0.281762, -0.114932, -0.243566,
       -0.188988,  0.007132,  0.146958, -0.269132, -0.034678,  0.164219,
       -0.090405,  0.044055, -0.405617,  0.162068,  0.008641,  0.142725,
       -0.065804,  0.136329,  0.247081, -0.144136, -0.007592,  0.113807,
        0.14899 , -0.012305, -0.346247,  0.111213,  0.011855,  0.250259,
        0.068319,  0.100238,  0.010446,  0.127471, -0.033004, -0.079141,
       -0.531317,  0.318433,  0.562701,  0.354442,  0.001789,  0.171425,
        0.010197,  0.043681, -0.031715,  0.505568, -0.453908,  0.050882,
       -0.294158, -0.279005, -0.121981,  0.182517,  0.419662,  0.325643,
       -0.027132, -0.020383,  0.133923, -0.047128,  0.406531, -0.385169,
       -0.148565, -0.109004, -0.062878, -0.236207,  0.008543,  0.044215,
        0.216761,  0.017197, -0.034889,  0.311272,  0.130243, -0.122846],
      dtype=float32)
```

Znajdziemy najbardziej podobne słowo do słowa kobieta.

In [9]:
```
wv.most_similar('kobieta')
```

Out[9]:
```
[('mężczyzna', 0.7043285369873047),
 ('dziewczyna', 0.6813511252403259),
 ('ciężarna', 0.612939715385437),
 ('dziewczynka', 0.6094770431518555),
 ('50-letnia', 0.5805984139442444),
 ('młoda', 0.5788260102272034),
 ('40-letnia', 0.5631040930747986),
 ('kobietą', 0.5611501336097717),
 ('dama', 0.5526077747344971),
 ('mężczyzną', 0.5519220232963562)]
```

Znajdź najbardziej podobne słowo do słowa "kobieta", które nie jest podobne do słowa "mężczyzna"

In [10]:
```
wv.most_similar(positive=['kobieta'], negative=['mężczyzna'])
```

Out[10]:
```
[('Teresa', 0.3597542345523834),
 ('Irlandka', 0.35389503836631775),
 ('Barbara', 0.33941537141799927),
 ('Helena', 0.33837634325027466),
 ('Eleonora', 0.32829549908638),
 ('mianowana', 0.3252837657928467),
 ('Maryse', 0.32269802689552307),
 ('Emmanuella', 0.32241636514663696),
 ('czarnoskóra', 0.3224010169506073),
 ('Izabela', 0.3222866654396057)]
```

W grupie "pies kot żubr budyń" znajdź słowo, które nie pasuje.

In [11]:
```
wv.doesnt_match(['pies', 'kot', 'żubr', 'budyń'])
```

Out[11]:
```
'budyń'
```

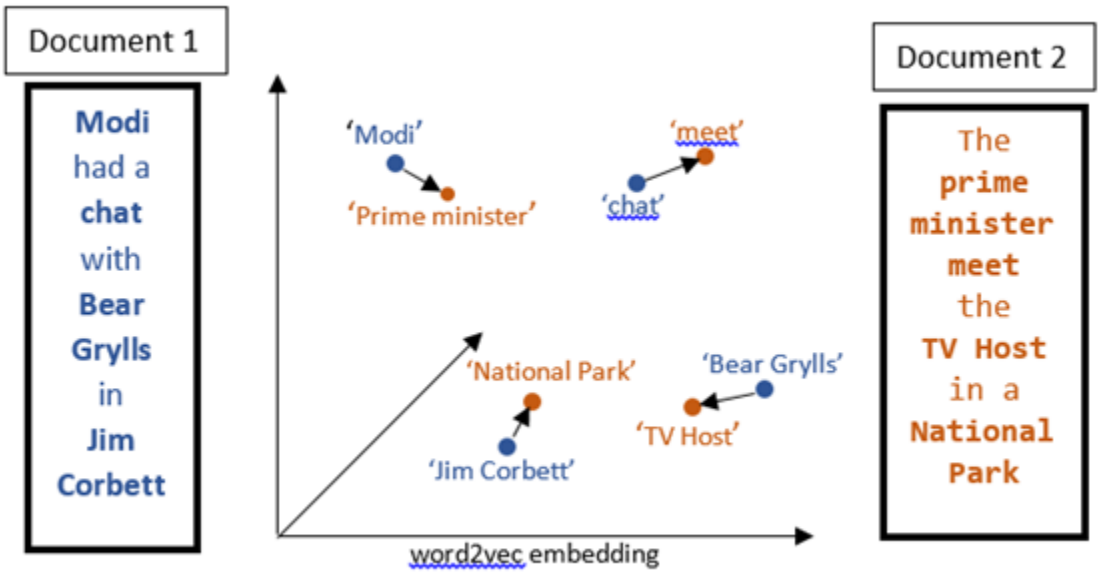
zbadaj podobieństwo między kotem a psem

In [12]:
```
wv.similarity('kot', 'pies')
```

Out[12]:
```
np.float32(0.6970217)
```

Teraz zajmiemy się podobieństwami pomiędzy zdaniami

Word mover distance to minimalna suma odległości, jaką trzeba „pokonać”, aby słowa z jednego zdania pokryły znaczeniowo słowa z drugiego zdania.



In [13]:
```
!pip install POT #Po tej instalacji gensim może szybciej i dokładniej liczyć wmdistance
```

```
Collecting POT
  Downloading pot-0.9.7.post1-cp313-cp313-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (44 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0.0/44.2 kB ? eta -:--:--
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 44.2/44.2 kB 3.0 MB/s eta
0:00:00
Requirement already satisfied: numpy⩾1.16 in
/usr/local/lib/python3.13/dist-packages (from POT) (2.1.3)
Requirement already satisfied: scipy⩾1.6 in /usr/local/lib/python3.13/dist-
packages (from POT) (1.16.3)
Downloading pot-0.9.7.post1-cp313-cp313-
manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (32.9 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 32.9/32.9 MB 22.2 MB/s eta
0:00:00
Installing collected packages: POT
Successfully installed POT-0.9.7.post1
```

In [14]:
```
sentence_1 = 'Byłem dzisiaj w kinie'.lower().split()
sentence_2 = 'Poszłam wczoraj do teatru'.lower().split()
```

In [15]:
```
wv.wmdistance(sentence_1, sentence_2)
```

Out[15]:
```
1.176677407878612
```

WMD nie jest ograniczone do 0-1, ale przy normalizacji embeddingów zwykle wartości mieszczą się w 0-4 dla krótkich zdań.

wynik ~1.25 sugeruje, że zdania są umiarkowanie podobne semantycznie.

Dla porównania:

- Dwa identyczne zdania → WMD ≈ 0
- Zupełnie różne zdania → WMD > 2.5-3

Flair

Flair to biblioteka NLP od Zalando Research (Python)

Główna idea:

- Klasyczne embeddingi (Word2Vec, GloVe, FastText) dają stały wektor dla każdego słowa, niezależnie od kontekstu:

  "bank" w *"river bank"* i *"money bank"* → ten sam wektor.

- Flair tworzy kontekstowe embeddingi(wektor słowa zależy od jego zdania / kontekstu)

  więc "*bank*" w "*river bank*" będzie inny niż w "*money bank*".

Flair uczy się tego przez modele LSTM działające w dwóch kierunkach (forward i backward).

In [1]:
```
!pip install flair
```

```
Collecting flair
  Downloading flair-0.15.1-py3-none-any.whl.metadata (12 kB)
Collecting boto3 ⩾ 1.20.27 (from flair)
  Downloading boto3-1.43.113-py3-none-any.whl.metadata (6.6 kB)
Collecting conllu<5.0.0,⩾4.0 (from flair)
  Downloading conllu-4.5.3-py2.py3-none-any.whl.metadata (19 kB)
Collecting deprecated ⩾ 1.2.13 (from flair)
  Downloading deprecated-3.0.0-py3-none-any.whl.metadata (6.5 kB)
Collecting ftfy ⩾ 6.1.0 (from flair)
  Downloading ftfy-6.3.1-py3-none-any.whl.metadata (7.3 kB)
Requirement already satisfied: gdown ⩾ 4.4.0 in
/usr/local/lib/python3.13/dist-packages (from flair) (5.2.2)
Requirement already satisfied: huggingface-hub ⩾ 0.10.0 in
/usr/local/lib/python3.13/dist-packages (from flair) (1.33.0)
Collecting langdetect ⩾ 1.0.9 (from flair)
  Downloading langdetect-1.0.9.tar.gz (981 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 981.5/981.5 kB 14.6 MB/s eta
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  Preparing metadata (setup.py) ... done
Requirement already satisfied: lxml ⩾ 4.8.0 in
/usr/local/lib/python3.13/dist-packages (from flair) (6.1.3)
Requirement already satisfied: matplotlib ⩾ 2.2.3 in
/usr/local/lib/python3.13/dist-packages (from flair) (3.10.0)
Requirement already satisfied: more-itertools ⩾ 8.13.0 in
/usr/local/lib/python3.13/dist-packages (from flair) (10.8.0)
Collecting mpld3 ⩾ 0.3 (from flair)
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Requirement already satisfied: python-dateutil ⩾ 2.8.2 in
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Collecting pytorch-revgrad ⩾ 0.2.0 (from flair)
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Requirement already satisfied: torch ⩾ 1.13.1 in
/usr/local/lib/python3.13/dist-packages (from flair) (2.11.0+cu130)
Requirement already satisfied: tqdm ⩾ 4.63.0 in
/usr/local/lib/python3.13/dist-packages (from flair) (4.67.3)
Collecting transformer-smaller-training-vocab ⩾ 0.2.3 (from flair)
  Downloading transformer_smaller_training_vocab-0.4.2-py3-none-
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any.whl.metadata (7.9 kB)
Collecting transformers<5.0.0,≥4.25.0 (from transformers[sentencepiece]
<5.0.0,≥4.25.0→flair)
  Downloading transformers-4.57.6-py3-none-any.whl.metadata (43 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 44.0/44.0 kB 4.3 MB/s eta
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Collecting botocore<1.44.0,≥1.43.113 (from boto3≥1.20.27→flair)
  Downloading botocore-1.43.113-py3-none-any.whl.metadata (5.6 kB)
Collecting jmespath<2.0.0,≥0.7.1 (from boto3≥1.20.27→flair)
  Downloading jmespath-1.1.0-py3-none-any.whl.metadata (7.6 kB)
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Requirement already satisfied: wrapt<3,≥1.16 in
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(2.5.0)
Requirement already satisfied: wcwidth in /usr/local/lib/python3.13/dist-
packages (from ftfy≥6.1.0→flair) (0.9.1)
Requirement already satisfied: beautifulsoup4 in
/usr/local/lib/python3.13/dist-packages (from gdown≥4.4.0→flair) (4.13.5)
Requirement already satisfied: filelock in /usr/local/lib/python3.13/dist-
packages (from gdown≥4.4.0→flair) (4.0.7)
Requirement already satisfied: requests[socks] in
/usr/local/lib/python3.13/dist-packages (from gdown≥4.4.0→flair) (2.32.4)
Requirement already satisfied: click<9.0.0,≥8.4.2 in
/usr/local/lib/python3.13/dist-packages (from huggingface-hub≥0.10.0
→flair) (8.5.0)
Requirement already satisfied: fsspec≥2023.5.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface-hub≥0.10.0
→flair) (2025.12.0)
Requirement already satisfied: hf-xet<2.0.0,≥1.6.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface-hub≥0.10.0
→flair) (1.6.0)
Requirement already satisfied: httpx<1,≥0.23.0 in
/usr/local/lib/python3.13/dist-packages (from huggingface-hub≥0.10.0
→flair) (0.28.1)
Requirement already satisfied: packaging≥20.9 in
/usr/local/lib/python3.13/dist-packages (from huggingface-hub≥0.10.0
→flair) (26.3)
Requirement already satisfied: pyyaml≥5.1 in
/usr/local/lib/python3.13/dist-packages (from huggingface-hub≥0.10.0
→flair) (6.0.3)
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 Requirement already satisfied: typing-extensions⩾4.1.0 in
 /usr/local/lib/python3.13/dist-packages (from huggingface-hub⩾0.10.0
→flair) (4.16.0)
 Requirement already satisfied: six in /usr/local/lib/python3.13/dist-
 packages (from langdetect⩾1.0.9→flair) (1.17.0)
 Requirement already satisfied: contourpy⩾1.0.1 in
 /usr/local/lib/python3.13/dist-packages (from matplotlib⩾2.2.3→flair)
 (1.4.0)
 Requirement already satisfied: cycler⩾0.10 in
 /usr/local/lib/python3.13/dist-packages (from matplotlib⩾2.2.3→flair)
 (0.12.1)
 Requirement already satisfied: fonttools⩾4.22.0 in
 /usr/local/lib/python3.13/dist-packages (from matplotlib⩾2.2.3→flair)
 (4.66.1)
 Requirement already satisfied: kiwisolver⩾1.3.1 in
 /usr/local/lib/python3.13/dist-packages (from matplotlib⩾2.2.3→flair)
 (1.5.1)
 Requirement already satisfied: numpy⩾1.23 in
 /usr/local/lib/python3.13/dist-packages (from matplotlib⩾2.2.3→flair)
 (2.1.3)
 Requirement already satisfied: pillow⩾8 in /usr/local/lib/python3.13/dist-
 packages (from matplotlib⩾2.2.3→flair) (11.3.0)
 Requirement already satisfied: pyparsing⩾2.3.1 in
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 Requirement already satisfied: jinja2 in /usr/local/lib/python3.13/dist-
 packages (from mpld3⩾0.3→flair) (3.1.6)
 Requirement already satisfied: scipy⩾1.6.0 in
 /usr/local/lib/python3.13/dist-packages (from scikit-learn⩾1.0.2→flair)
 (1.16.3)
 Requirement already satisfied: joblib⩾1.2.0 in
 /usr/local/lib/python3.13/dist-packages (from scikit-learn⩾1.0.2→flair)
 (1.6.0)
 Requirement already satisfied: threadpoolctl⩾3.1.0 in
 /usr/local/lib/python3.13/dist-packages (from scikit-learn⩾1.0.2→flair)
 (3.7.0)
 Requirement already satisfied: setuptools<82 in
 /usr/local/lib/python3.13/dist-packages (from torch⩾1.13.1→flair)
 (80.10.2)
 Requirement already satisfied: sympy⩾1.13.3 in
 /usr/local/lib/python3.13/dist-packages (from torch⩾1.13.1→flair) (1.14.0)
 Requirement already satisfied: networkx⩾2.5.1 in
 /usr/local/lib/python3.13/dist-packages (from torch⩾1.13.1→flair) (3.7)
 Requirement already satisfied: cuda-toolkit==13.0.2 in
 /usr/local/lib/python3.13/dist-packages (from cuda-
 toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
 nvrtc,nvtx]==13.0.2; platform_system == "Linux"→torch⩾1.13.1→flair)
 (13.0.2)
 Requirement already satisfied: cuda-bindings<14,⩾13.0.3 in
 /usr/local/lib/python3.13/dist-packages (from torch⩾1.13.1→flair) (13.0.3)
 Requirement already satisfied: nvidia-cudnn-cu13==9.19.0.56 in
 /usr/local/lib/python3.13/dist-packages (from torch⩾1.13.1→flair)
 (9.19.0.56)
 Requirement already satisfied: nvidia-cusparselt-cu13==0.8.0 in
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/usr/local/lib/python3.13/dist-packages (from torch⩾1.13.1→flair) (0.8.0)
Requirement already satisfied: nvidia-nccl-cu13==2.28.9 in
/usr/local/lib/python3.13/dist-packages (from torch⩾1.13.1→flair) (2.28.9)
Requirement already satisfied: nvidia-nvshmem-cu13==3.4.5 in
/usr/local/lib/python3.13/dist-packages (from torch⩾1.13.1→flair) (3.4.5)
Requirement already satisfied: triton==3.6.0 in
/usr/local/lib/python3.13/dist-packages (from torch⩾1.13.1→flair) (3.6.0)
Requirement already satisfied: nvidia-cublas==13.1.0.3.* in
/usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
nvrtc,nvtx]==13.0.2; platform_system == "Linux"→torch⩾1.13.1→flair)
(13.1.0.3)
Requirement already satisfied: nvidia-cuda-runtime==13.0.96.* in
/usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
nvrtc,nvtx]==13.0.2; platform_system == "Linux"→torch⩾1.13.1→flair)
(13.0.96)
Requirement already satisfied: nvidia-cufft==12.0.0.61.* in
/usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
nvrtc,nvtx]==13.0.2; platform_system == "Linux"→torch⩾1.13.1→flair)
(12.0.0.61)
Requirement already satisfied: nvidia-cufile==1.15.1.6.* in
/usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
nvrtc,nvtx]==13.0.2; platform_system == "Linux"→torch⩾1.13.1→flair)
(1.15.1.6)
Requirement already satisfied: nvidia-cuda-cupti==13.0.85.* in
/usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
nvrtc,nvtx]==13.0.2; platform_system == "Linux"→torch⩾1.13.1→flair)
(13.0.85)
Requirement already satisfied: nvidia-curand==10.4.0.35.* in
/usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
nvrtc,nvtx]==13.0.2; platform_system == "Linux"→torch⩾1.13.1→flair)
(10.4.0.35)
Requirement already satisfied: nvidia-cusolver==12.0.4.66.* in
/usr/local/lib/python3.13/dist-packages (from cuda-
toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
nvrtc,nvtx]==13.0.2; platform_system == "Linux"→torch⩾1.13.1→flair)
(12.0.4.66)
Requirement already satisfied: nvidia-cusparse==12.6.3.3.* in
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nvrtc,nvtx]==13.0.2; platform_system == "Linux"→torch⩾1.13.1→flair)
(12.6.3.3)
Requirement already satisfied: nvidia-nvjitlink==13.0.88.* in
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toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
nvrtc,nvtx]==13.0.2; platform_system == "Linux"→torch⩾1.13.1→flair)
(13.0.88)
Requirement already satisfied: nvidia-cuda-nvrtc==13.0.88.* in
/usr/local/lib/python3.13/dist-packages (from cuda-
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 toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparse,nvjitlink,
 nvrtc,nvtx]==13.0.2; platform_system == "Linux"→torch≥1.13.1→flair)
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 (13.0.85)
 Collecting huggingface-hub≥0.10.0 (from flair)
   Downloading huggingface_hub-0.36.2-py3-none-any.whl.metadata (15 kB)
 Collecting tokenizers≤0.23.0,≥0.22.0 (from transformers<5.0.0,≥4.25.0
→transformers[sentencepiece]<5.0.0,≥4.25.0→flair)
   Downloading tokenizers-0.22.2-cp39-abi3-
 manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (7.3 kB)
 Requirement already satisfied: safetensors≥0.4.3 in
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→transformers[sentencepiece]<5.0.0,≥4.25.0→flair) (0.8.0)
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 <5.0.0,≥4.25.0→flair) (0.2.2)
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 Requirement already satisfied: tenacity==9.1.4 in
 /usr/local/lib/python3.13/dist-packages (from wikipedia-api≥0.5.7→flair)
 (9.1.4)
 Requirement already satisfied: anyio in /usr/local/lib/python3.13/dist-
 packages (from httpx<1,≥0.23.0→huggingface-hub≥0.10.0→flair) (4.15.1)
 Requirement already satisfied: certifi in /usr/local/lib/python3.13/dist-
 packages (from httpx<1,≥0.23.0→huggingface-hub≥0.10.0→flair) (2026.7.22)
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 hub≥0.10.0→flair) (1.0.9)
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→flair) (0.16.0)
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 /usr/local/lib/python3.13/dist-packages (from botocore<1.44.0,≥1.43.113
→boto3≥1.20.27→flair) (2.5.0)
 Requirement already satisfied: cuda-pathfinder~=1.1 in
 /usr/local/lib/python3.13/dist-packages (from cuda-bindings<14,≥13.0.3
→torch≥1.13.1→flair) (1.8.2)
 Requirement already satisfied: cloudpickle≥3.0 in
 /usr/local/lib/python3.13/dist-packages (from joblib≥1.2.0→scikit-
 learn≥1.0.2→flair) (3.1.2)
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→bioc<3.0.0,≥2.0.0→flair) (26.1.0)
 Requirement already satisfied: mpmath<1.4,≥1.1.0 in
 /usr/local/lib/python3.13/dist-packages (from sympy≥1.13.3→torch≥1.13.1
→flair) (1.3.0)
 Requirement already satisfied: accelerate≥0.26.0 in
 /usr/local/lib/python3.13/dist-packages (from
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 transformers[sentencepiece,torch]<5.0,⩾4.1→transformer-smaller-training-
 vocab⩾0.2.3→flair) (1.15.0)
 Requirement already satisfied: soupsieve>1.2 in
 /usr/local/lib/python3.13/dist-packages (from beautifulsoup4→gdown⩾4.4.0
→flair) (2.10)
 Requirement already satisfied: sortedcontainers in
 /usr/local/lib/python3.13/dist-packages (from intervaltree
→bioc<3.0.0,⩾2.0.0→flair) (2.4.0)
 Requirement already satisfied: MarkupSafe⩾2.0 in
 /usr/local/lib/python3.13/dist-packages (from jinja2→mpld3⩾0.3→flair)
 (3.0.3)
 Requirement already satisfied: charset_normalizer<4,⩾2 in
 /usr/local/lib/python3.13/dist-packages (from requests[socks]→gdown⩾4.4.0
→flair) (3.4.9)
 Requirement already satisfied: PySocks≠1.5.7,⩾1.5.6 in
 /usr/local/lib/python3.13/dist-packages (from requests[socks]→gdown⩾4.4.0
→flair) (1.7.1)
 Requirement already satisfied: psutil in /usr/local/lib/python3.13/dist-
 packages (from accelerate⩾0.26.0→transformers[sentencepiece,torch]
 <5.0,⩾4.1→transformer-smaller-training-vocab⩾0.2.3→flair) (5.9.5)
 Downloading flair-0.15.1-py3-none-any.whl (1.2 MB)
    ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.2/1.2 MB 58.3 MB/s eta 0:00:00
 Downloading bioc-2.1-py3-none-any.whl (33 kB)
 Downloading boto3-1.43.113-py3-none-any.whl (140 kB)
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 Downloading conllu-4.5.3-py2.py3-none-any.whl (16 kB)
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 Downloading ftfy-6.3.1-py3-none-any.whl (44 kB)
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 Downloading pytorch_revgrad-0.2.0-py3-none-any.whl (4.6 kB)
 Downloading segtok-1.5.11-py3-none-any.whl (24 kB)
 Downloading transformer_smaller_training_vocab-0.4.2-py3-none-any.whl (14
 kB)
 Downloading transformers-4.57.6-py3-none-any.whl (12.0 MB)
    ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.0/12.0 MB 112.6 MB/s eta
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 Downloading huggingface_hub-0.36.2-py3-none-any.whl (566 kB)
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 0:00:00
 Downloading jmespath-1.1.0-py3-none-any.whl (20 kB)
 Downloading jsonlines-4.0.0-py3-none-any.whl (8.7 kB)
 Downloading s3transfer-0.19.2-py3-none-any.whl (90 kB)
    ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 90.2/90.2 kB 10.2 MB/s eta
```

```
0:00:00
Downloading tokenizers-0.22.2-cp39-abi3-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl (3.3 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.3/3.3 MB 114.3 MB/s eta
0:00:00
Downloading intervaltree-3.2.1-py2.py3-none-any.whl (25 kB)
Building wheels for collected packages: langdetect, pptree, sqlitedict,
docopt
  Building wheel for langdetect (setup.py) ... done
  Created wheel for langdetect: filename=langdetect-1.0.9-py3-none-any.whl
size=993335
sha256=0ac778e4902b1f3bfeb47412ac1180a5d3fa95f8d1dbb387caedbf5a10abcdf9
  Stored in directory:
/root/.cache/pip/wheels/eb/87/25/2dddf1c94e1786054e25022ec5530bfed52bad86d88
2999c48
  Building wheel for pptree (setup.py) ... done
  Created wheel for pptree: filename=pptree-3.1-py3-none-any.whl size=4641
sha256=b1bd0b5b4c8e2f5eb4034b2a5a889b43537f700e9d7cabb209cb315e7585c1bb
  Stored in directory:
/root/.cache/pip/wheels/97/5c/d1/a8347c9c986fc222c12240fafdde88e50a72553e19d
7517177
  Building wheel for sqlitedict (setup.py) ... done
  Created wheel for sqlitedict: filename=sqlitedict-2.1.0-py3-none-any.whl
size=16958
sha256=d0ce04439d8672312f0124885ca77ff52606b1f38c75e0e8a063a0eb112f8f59
  Stored in directory:
/root/.cache/pip/wheels/56/1e/68/fc282426da812281e81368e59ed19f3edc6d9eef451
8dbc41c
  Building wheel for docopt (setup.py) ... done
  Created wheel for docopt: filename=docopt-0.6.2-py2.py3-none-any.whl
size=13781
sha256=adc559d9d79a7cc9d19608790b12dc8ab353fd6ebe34809477258321ca2696d2
  Stored in directory:
/root/.cache/pip/wheels/0b/1d/03/175286677fb5a1341cc3e4755bf8ec0ed08f3329afd
67446b0
Successfully built langdetect pptree sqlitedict docopt
Installing collected packages: sqlitedict, pptree, docopt, segtok,
langdetect, jsonlines, jmespath, intervaltree, ftfy, deprecated, conllu,
huggingface-hub, botocore, bioc, wikipedia-api, tokenizers, s3transfer,
mpld3, transformers, boto3, pytorch-revgrad, transformer-smaller-training-
vocab, flair
  Attempting uninstall: huggingface-hub
    Found existing installation: huggingface_hub 1.33.0
    Uninstalling huggingface_hub-1.33.0:
      Successfully uninstalled huggingface_hub-1.33.0
  Attempting uninstall: tokenizers
    Found existing installation: tokenizers 0.23.2
    Uninstalling tokenizers-0.23.2:
      Successfully uninstalled tokenizers-0.23.2
  Attempting uninstall: transformers
    Found existing installation: transformers 5.18.0
    Uninstalling transformers-5.18.0:
      Successfully uninstalled transformers-5.18.0
ERROR: pip's dependency resolver does not currently take into account all
```

```
the packages that are installed. This behaviour is the source of the
following dependency conflicts.
gradio 6.29.0 requires huggingface-hub<3.0,≥1.16.0, but you have
huggingface-hub 0.36.2 which is incompatible.
diffusers 0.40.0 requires huggingface-hub<2.0,≥1.23.0, but you have
huggingface-hub 0.36.2 which is incompatible.
Successfully installed bioc-2.1 boto3-1.43.113 botocore-1.43.113 conllu-
4.5.3 deprecated-3.0.0 docopt-0.6.2 flair-0.15.1 ftfy-6.3.1 huggingface-hub-
0.36.2 intervaltree-3.2.1 jmespath-1.1.0 jsonlines-4.0.0 langdetect-1.0.9
mpld3-0.5.12 pptree-3.1 pytorch-revgrad-0.2.0 s3transfer-0.19.2 segtok-
1.5.11 sqlitedict-2.1.0 tokenizers-0.22.2 transformer-smaller-training-
vocab-0.4.2 transformers-4.57.6 wikipedia-api-0.16.0
```

In [2]:

```
from flair.embeddings import FlairEmbeddings, StackedEmbeddings
from flair.data import Sentence
```

In [3]:

```
embeddings = FlairEmbeddings('news-forward')

#tworzymy objekty Sentence
s1 = Sentence("How does it feel to be a scribe?")
s2 = Sentence("There is no such thing as simply good or bad")
sentences = [s1, s2]


# Obliczamy embeddingi(zanurzamy słowa)
embeddings.embed(sentences)

for sentence in sentences: #Zanurzone zdania nadal są obiektem Sentence
  print(sentence)
```

```
2026-10-14 16:36:30,933 https://flair.informatik.hu-
berlin.de/resources/embeddings/flair/news-forward-0.4.1.pt not found in
cache, downloading to /tmp/tmpyc5ar69c
```

```
100%|██████████| 69.7M/69.7M [00:04<00:00, 15.2MB/s]
```

```
2026-10-14 16:36:36,220 copying /tmp/tmpyc5ar69c to cache at
/root/.flair/embeddings/news-forward-0.4.1.pt
2026-10-14 16:36:36,269 removing temp file /tmp/tmpyc5ar69c
```

```

```

```
Sentence[9]: "How does it feel to be a scribe?"
Sentence[10]: "There is no such thing as simply good or bad"
```

In [4]:
```
#dostajemy embeddingi poszczególnych słów
for sentence in sentences:
  for token in sentence:
    print(token.embedding)
#embeddingi są tensorami
```

```
tensor([-0.0011, -0.0021,  0.0038,  ..., -0.0004,  0.0099,  0.0547],
       device='cuda:0')
tensor([-0.0016, -0.0260,  0.0388,  ...,  0.0031, -0.0070,  0.0025],
       device='cuda:0')
tensor([0.0174, 0.0181, 0.0148,  ..., 0.0051, 0.0051, 0.0130],
device='cuda:0')
tensor([ 0.0001,  0.0128,  0.0476,  ..., -0.0084, -0.0023,  0.0106],
       device='cuda:0')
tensor([-2.1672e-03,  8.3752e-05,  2.5102e-02,  ..., -2.8347e-04,
        -1.2090e-02,  2.9513e-02], device='cuda:0')
tensor([-0.0032,  0.0020,  0.0280,  ...,  0.0004, -0.0017, -0.0106],
       device='cuda:0')
tensor([ 5.3283e-03,  3.8049e-05, -2.2042e-02,  ..., -4.1646e-04,
         1.9279e-02, -1.2351e-02], device='cuda:0')
tensor([-1.0594e-03,  4.8670e-05,  2.1462e-02,  ..., -1.1294e-05,
        -4.3308e-03,  3.1280e-03], device='cuda:0')
tensor([ 8.7317e-04,  3.7862e-06, -4.6364e-04,  ..., -1.4222e-04,
         5.4708e-03,  3.0888e-03], device='cuda:0')
tensor([-0.0012, -0.0002, -0.1499,  ...,  0.0024,  0.0017,  0.1648],
       device='cuda:0')
tensor([ 0.0078, -0.0047,  0.0564,  ..., -0.0012,  0.0325,  0.0035],
       device='cuda:0')
tensor([ 5.8073e-02, -7.9190e-05,  9.8895e-02,  ..., -2.7518e-03,
         8.8327e-04,  2.6957e-02], device='cuda:0')
tensor([-0.0004, -0.0036,  0.0781,  ..., -0.0004, -0.0344, -0.0005],
       device='cuda:0')
tensor([ 0.0014,  0.0033, -0.0118,  ..., -0.0005, -0.0164,  0.0293],
       device='cuda:0')
tensor([ 1.2536e-02, -1.0569e-03, -1.1876e-02,  ..., -2.9878e-05,
         4.0988e-03,  2.8744e-02], device='cuda:0')
tensor([-0.0053, -0.0011,  0.0058,  ..., -0.0018,  0.0212,  0.0082],
       device='cuda:0')
tensor([ 2.0060e-04,  9.6346e-04, -2.2168e-02,  ...,  6.8233e-05,
        -1.3982e-01,  3.5051e-03], device='cuda:0')
tensor([-9.0028e-05, -1.7683e-03,  1.0521e-02,  ..., -1.9229e-03,
        -1.2467e-05,  3.2030e-02], device='cuda:0')
tensor([-0.0006,  0.0004, -0.0019,  ..., -0.0005,  0.0061, -0.0073],
       device='cuda:0')
```

In [5]:
```
import numpy as np
```

In [7]:
```
for sentence in sentences:
  for token in sentence:
    print(np.array(token.embedding))
```

```
/tmp/ipykernel_432/932037997.py:3: DeprecationWarning: __array__
implementation doesn't accept a copy keyword, so passing copy=False failed.
__array__ must implement 'dtype' and 'copy' keyword arguments.
  print(np.array(token.embedding))
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipykernel_432/932037997.py in <cell line: 0>()
      1 for sentence in sentences:
      2   for token in sentence:
----> 3     print(np.array(token.embedding))

/usr/local/lib/python3.13/dist-packages/torch/_tensor.py in __array__(self,
dtype)
   1251             return handle_torch_function(Tensor.__array__, (self,),
self, dtype=dtype)
   1252         if dtype is None:
→ 1253             return self.numpy()
   1254         else:
   1255             return self.numpy().astype(dtype, copy=False)

TypeError: can't convert cuda:0 device type tensor to numpy. Use
Tensor.cpu() to copy the tensor to host memory first.
```

Embeddingi dla zdań

In [8]:
```
from flair.embeddings import DocumentPoolEmbeddings
```

In [9]:
```
sentence_embeddings = DocumentPoolEmbeddings([FlairEmbeddings('news-
forward')])
```

In [10]:
```
sentence_embeddings.embed(sentences)
```

In [11]:
```
for sentence in sentences:
  print(sentence.embedding)
```

```
tensor([ 0.0016,  0.0005,  0.0175,  ..., -0.0001,  0.0014,  0.0104],
       device='cuda:0')
tensor([ 0.0073, -0.0008,  0.0052,  ..., -0.0007, -0.0124,  0.0289],
       device='cuda:0')
```

Zadania

###Zadanie1

Załaduj polski model kontekstowych embeddingów Flair:

In [12]:
```
embeddings = FlairEmbeddings('pl-forward')
```

```
2026-10-14 16:37:27,589 https://flair.informatik.hu-
berlin.de/resources/embeddings/flair/lm-polish-forward-v0.2.pt not found in
cache, downloading to /tmp/tmpwy5ntxn6
```

```
100%|██████████| 80.3M/80.3M [00:05<00:00, 16.0MB/s]
```

```
2026-10-14 16:37:34,998 copying /tmp/tmpwy5ntxn6 to cache at
/root/.flair/embeddings/lm-polish-forward-v0.2.pt
2026-10-14 16:37:35,058 removing temp file /tmp/tmpwy5ntxn6
```

In [13]:
```
s1 = Sentence("Zamek w Malborku jest bardzo stary.")
s2 = Sentence("Suwak w kurtce to zamek błyskawiczny")
s3 = Sentence("Sprawdź, czy zamek w drzwiach działa")
sentences = [s1, s2, s3]
```

Oblicz embeddingi dla wszystkich zdań:

In [14]:
```
embeddings.embed(sentences)
```

Out[14]:
```
[Sentence[7]: "Zamek w Malborku jest bardzo stary.",
 Sentence[6]: "Suwak w kurtce to zamek błyskawiczny",
 Sentence[7]: "Sprawdź, czy zamek w drzwiach działa"]
```

Wypisz każde zdanie i dla słowa "zamek", wypisz pierwsze 5 wymiarów embeddingu:

In [15]:
```
for sentence in sentences:
    print(sentence)
    for token in sentence:
        if token.text.lower() == "zamek":
            print("  zamek →", token.embedding[:5])
```

```
Sentence[7]: "Zamek w Malborku jest bardzo stary."
  zamek → tensor([-5.4509e-03, -2.1590e-02,  4.5457e-05, -1.7073e-04,
-4.3133e-04],
       device='cuda:0')
Sentence[6]: "Suwak w kurtce to zamek błyskawiczny"
  zamek → tensor([-0.0043, -0.0131,  0.0004, -0.0001, -0.0004],
device='cuda:0')
Sentence[7]: "Sprawdź, czy zamek w drzwiach działa"
  zamek → tensor([-4.3021e-03, -2.1082e-02,  7.7413e-04, -6.9932e-05,
-3.6798e-04],
       device='cuda:0')
```

Czy embeddingi są różne w zależności od znaczenia słowa?

Odpowiedź: Tak

###Zadanie 2

Stwórz bota z bazą przykładownych pytań i odpowiedzi, zwracającego odpowiedź na zapytanie na podstawie podobieństwa (WMD albo flair document embeddings) pomiędzy zapytniem użytownika a pytaniami w swojej bazie

Przykład:

Zbiór pytań i odpowiedzi :

```
[
  {"q1" : "a1"},
  {"q2" : "a2"},
  {"q3" : "a3"},
  ...
  {"qn" : "an"},
]
```

Wektoryzujemy pytania:

```
  {"q1_vecotorized" : "a1"},
  {"q2_vecotorized" : "a2"},
  {"q3_vecotorized" : "a3"},
  ...
  {"qn_vecotorized" : "an"},
```

Dla pytania użytkownia pytania q_user wektoryzujemy je i liczmy podobieństwo z pytaniami w bazie

```
sim("q_user_vectorized","q1_vecotorized"), sim("q_user_vectorized","q2_vecotorized"),
... ("q_user_vectorized""qn_vecotorized")
```

Zwracamy odpowiedź do pytania z bazy najbardziej podobnego do pytania użytkownika

$$argmax(sim(q_{user}vectorized, q_ivecotorized))$$

Do liczenia podobnieństma można użyć np https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.distance.cosine.html ale wtedy robimy argmin bo

$$cosine\ distance = 1 - cosine\ similarity$$

In [17]:
```
qa = [
    {"q": "O której otwieracie sklep?", "a": "Sklep jest otwarty od 9:00 do
21:00."},
    {"q": "Czy mogę zapłacić kartą?", "a": "Tak, przyjmujemy karty i
BLIK."},
    {"q": "Ile kosztuje dostawa?", "a": "Dostawa kosztuje 15 zł."},
    {"q": "Czy mogę zwrócić towar?", "a": "Tak, masz 14 dni na zwrot."},
    {"q": "Gdzie znajduje się sklep?", "a": "Sklep jest w Warszawie przy
ulicy Długiej 5."},
]
```

In [18]:
```python
# importy
from scipy.spatial.distance import cosine
from flair.data import Sentence
from flair.embeddings import FlairEmbeddings, DocumentPoolEmbeddings

doc_embeddings = DocumentPoolEmbeddings([FlairEmbeddings('pl-forward')])

def wektor(tekst):
    s = Sentence(tekst)
    doc_embeddings.embed(s)
    return s.embedding.cpu().detach().numpy()

def bot(user_question):
    user_vec = wektor(user_question)

    najlepsze_sim = -1
    najlepsze = None
    for para in qa:
        sim = calculate_similarity(user_vec, wektor(para["q"]))
        if sim > najlepsze_sim:
            najlepsze_sim = sim
            najlepsze = para

    print(f"Pytanie użytkownika: {user_question}")
    print(f"Najbardziej podobne pytanie z bazy danych: {najlepsze['q']}")
    print(f"Odpowiedź: {najlepsze['a']}")

def calculate_similarity(vec1, vec2):
    return 1 - cosine(vec1, vec2)
```

In [19]:
```python
# TODO - zadaj 3 pytania
bot("Do której jest otwarte?")
bot("Czy można płacić kartą?")
bot("Ile płacę za dostawę?")
```

```
Pytanie użytkownika: Do której jest otwarte?
Najbardziej podobne pytanie z bazy danych: O której otwieracie sklep?
Odpowiedź: Sklep jest otwarty od 9:00 do 21:00.
Pytanie użytkownika: Czy można płacić kartą?
Najbardziej podobne pytanie z bazy danych: Czy mogę zapłacić kartą?
Odpowiedź: Tak, przyjmujemy karty i BLIK.
Pytanie użytkownika: Ile płacę za dostawę?
Najbardziej podobne pytanie z bazy danych: Ile kosztuje dostawa?
Odpowiedź: Dostawa kosztuje 15 zł.
```